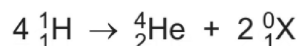


10 (a) Define the binding energy of a nucleus.

.....
.....
..... [2]

(b) A star emits radiation that is produced in a series of nuclear reactions that form helium from hydrogen. The reactions may be summarised by the overall equation



where X is a particle that has no binding energy.

All of the radiation emitted by the star is produced by this series of nuclear reactions.

The mass defect of the ${}_2^4\text{He}$ nuclide is $0.030\,377\,\text{u}$, where u is the unified atomic mass unit.

(i) State the name of particle X.

..... [1]

(ii) Show that the binding energy of the ${}_2^4\text{He}$ nuclide is $4.54 \times 10^{-12}\,\text{J}$.

[2]

(c) The star in **10(b)** has a luminosity of $7.98 \times 10^{27}\,\text{W}$. An observer on the Earth detects radiation from this star at a distance away of $8.15 \times 10^{16}\,\text{m}$.

(i) Calculate the radiant flux intensity F detected by the observer.

$F = \dots\dots\dots \text{Wm}^{-2}$ [2]

- (ii) The mass of hydrogen in the star decreases continuously as a result of the series of nuclear reactions.

Determine the rate of decrease of mass of hydrogen in the star.

rate of decrease of mass = kg s^{-1} [4]

[Total: 11]